

GenMol UDLM engineering exploration

followthrough-resolution-20260907-r1 | complete | 2026-09-07T11:43:08.191665+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

Prospective training settings: initialization: Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.; optimizer updates per arm: 1000; global batch: 128; training seed: 1500; training GPUs: 2; requested example exposures per arm: 128000; common clean-target mask: All tokenizer special IDs immutable clean targets; full active corruption vocabulary.. The MDLM and paper baselines remain contextual comparisons.

CT and clean CE share the training settings recorded in this protocol. Equal updates and examples do not imply equal compute or independently optimized objectives; training losses have different scales and targets.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED $\geq$ 0.6 and SA $\leq$ 4. Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Prospective sampling design

{`"corrector_steps_per_molecule": 0, "inference_eps": 1e-05, "inference_weights": "ema", "paired_contrasts": "512 minus 128 separately for CT and CE, paired by seed; both signs retained.", "primary_endpoint": "strict quality", "raw_loo_top_p": 1.0, "secondary_endpoints": ["strict validity", "released quality", "released validity", "uniqueness", "diversity", "odd ring labels", "unbalanced parentheses", "wall time"], "selection_history": "V9 primary CT/CE and CT temperature-0.5 summaries were inspected before specification. No follow-through seed or 512-step molecular output was inspected.", "softmax_temp": 1.0}`

512 predictor calls cost four times 128 calls; improvement is not an equal-compute win.

Two generation seeds do not measure variability over independent training runs.

The shared empirical categorical prior is a UDLM-derived variant, not strictly uniform noise.

A learning-curve experiment and held-out confirmation remain conditional on these outcomes.

CT and CE share hyperparameters selected using CT; this does not establish individually optimized objectives.

Objective comparison

CT / raw-LOO predicts clean leave-one-out probabilities. Clean CE predicts clean denoiser probabilities, converted to LOO logits at the current noisy state and time before temperature or top-p. The conversion adds no backbone evaluation. Shared sampling settings do not imply independently optimized objectives. Objective labels and checkpoint evidence below describe completed runs only; the prospective design also covers pending runs.

Training provenance: {"accumulate_grad_batches": 4, "arms": {"CE": {"checkpoint_sha256": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "controller_status": "completed", "launch_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/launch_manifest.json", "sha256": "903819961be96f71da79774d723626d8c4eb1aa50bf482d2a644c7a0dda22315", "size_bytes": 17020}, "protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8b_ce_followup.json", "sha256": "acb286ac4b938aa049327740affd2dea5beb0ded745cd51dca627f5e8fd9de8b"}, "request_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/request_manifest.json", "sha256": "c05fafdd1472e894b553afe580b051b1b2eec4c5dd47f7323df2b939e51f958d", "size_bytes": 8621}, "resolved_config_sha256": "80bd9866ef157690e0ef9ff553b0346b3b91f989c0193d0484c4997ac089b2bb", "source_revision": "b84f21ba1f30ceb62d2626daf2caad8adb6534cf", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8b/ce_e_1000_b128_w2/terminal_manifest.json", "sha256": "2dd0423257e8908548b69a28b8aa24b3cc956507944c343e24ef615253af2205", "size_bytes": 11719}, "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}, "CT": {"checkpoint_sha256": "48986899c401c09cdc1e9e865e773cadcd40899b62129420689f89a8e622a9c99", "checkpoint_status": "separately_validated_after_controller_failure", "controller_status": "failed", "post_exit_audit": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/post_exit_audit/post_exit_audit.json", "sha256": "959151c37072431be23b1f5696d7215da5291e0bace40d6f307864a7262f93c2", "size_bytes": 9166}, "resolved_config_sha256": "adbf518da62ea14d0e0b2e11f1ea91e058b25d0d35a59c3e8edfef060b5f0c46", "source_revision": "c8434dda105c5fb2a15bb784d24e0387062c733e", "terminal_receipt": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/ct/terminal_manifest.json", "sha256": "1021ba513564a25ca26ea3c2d6c148d271e1cd98d27078056285e848c9c69b5e", "size_bytes": 9822}}}, "common_mask_policy": "All tokenizer special IDs immutable clean targets; full active corruption vocabulary.", "conditioning_variant": "film_adaln", "example_exposures_per_arm": 128000, "global_batch_size": 128, "gpu_count": 2, "initialization": "Fresh common MDLM 50,000-update EMA; fresh optimizer and EMA for both arms.", "matched_common_config_sha256": "f373121441fca1aaa2f2657a76d7f7ce66be199cd35637597d0df240e257e89a", "micro_batch_size": 16, "optimizer_updates": 1000, "original_failed_campaign": {"relative_path": "output/udlm/engineering_v8/ct_ce_e_1000_b128_w2/campaign_terminal.json", "sha256": "98075a11c7340dafcc855887348201477981bd2a1a8f5d39053da24866cc663b", "size_bytes": 846}, "original_v8_protocol": {"relative_path": "experiments/udlm/protocols/engineering_v8_objectives.json", "sha256": "a27875752d25a4a7928401434f9bd90ee6e9d01539586bab70ce3ba75499026e"}, "peak_learning_rate": 0.0003, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "scheduler": {"decay_floor_lr": 3e-06, "horizon_updates": 1000, "name": "half_cosine_with_linear_warmup_and_floor", "warmup_updates": 50}, "training_implementation_hashes_sha256": "c9641b7e447d1f4617599e4df338fde0010fe68f7877566de37d2e6c5a6ccd2b", "training_seed": 1500, "uniform_mixture": 0.0002}

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdmlm_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
ct_n128	17300	128	128	0

Config	Seed	Total NFE	Predictors	Correctors
ct_n128	17301	128	128	0
ct_n512	17300	512	512	0
ct_n512	17301	512	512	0
ce_n128	17300	128	128	0
ce_n128	17301	128	128	0
ce_n512	17300	512	512	0
ce_n512	17301	512	512	0

Checkpoint logit interpretation

Config	Seed	Objective	Parameterization	Weights
ct_n128	17300	CT / raw-LOO	raw_loo	ema
ct_n128	17301	CT / raw-LOO	raw_loo	ema
ct_n512	17300	CT / raw-LOO	raw_loo	ema
ct_n512	17301	CT / raw-LOO	raw_loo	ema
ce_n128	17300	clean CE	x0_denoiser	ema
ce_n128	17301	clean CE	x0_denoiser	ema
ce_n512	17300	clean CE	x0_denoiser	ema
ce_n512	17301	clean CE	x0_denoiser	ema

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_n128	2/2	0.9750 +/- 0.0212	0.9949 +/- 0.0071	0.4400 +/- 0.0424	0.9042 +/- 0.0068
ct_n512	2/2	0.9550 +/- 0.0071	1.0000 +/- 0.0000	0.4300 +/- 0.0141	0.9034 +/- 0.0098

Config	Seeds complete	validity	uniqueness	quality	diversity
ce_n128	2/2	0.9850 +/- 0.0071	0.9949 +/- 0.0071	0.4050 +/- 0.0071	0.8979 +/- 0.0079
ce_n512	2/2	0.9750 +/- 0.0071	1.0000 +/- 0.0000	0.4400 +/- 0.0141	0.9027 +/- 0.0037

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
ct_n128	2/2	0.5450 +/- 0.0495	1.0000 +/- 0.0000	0.2850 +/- 0.0354	0.8908 +/- 0.0061
ct_n512	2/2	0.5300 +/- 0.0000	1.0000 +/- 0.0000	0.2550 +/- 0.0212	0.8801 +/- 0.0072
ce_n128	2/2	0.5800 +/- 0.0424	1.0000 +/- 0.0000	0.2800 +/- 0.0283	0.8815 +/- 0.0025
ce_n512	2/2	0.6200 +/- 0.0141	1.0000 +/- 0.0000	0.3000 +/- 0.0141	0.8878 +/- 0.0005

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
ct_n128	17300	completed	100	100	0.4700	0.3100	13.78
ct_n128	17301	completed	100	100	0.4100	0.2600	13.32
ct_n512	17300	completed	100	100	0.4200	0.2400	54.10
ct_n512	17301	completed	100	100	0.4400	0.2700	43.75
ce_n128	17300	completed	100	100	0.4100	0.3000	14.02
ce_n128	17301	completed	100	100	0.4000	0.2600	13.68
ce_n512	17300	completed	100	100	0.4300	0.3100	54.76
ce_n512	17301	completed	100	100	0.4500	0.2900	53.13

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path':

/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path':
'/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} -
config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected
'3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: 6976a1e299674bfb82fb8528c8832608379de18619fbda41aa3d12a19ae717cf

followthrough-ct-n128-r1 / seed 17300: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}, {"pid": 505542, "process_name": "[No data]", "used_memory_mib": 3134}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 4447, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ct-n128-r1/seed_17300/raw_samples.csv", "sha256": "628686b16e83886a7922f6a9e8983cfd4aa45bb2ec33cd113461121c8245506f", "size_bytes": 32297}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "6984cf8143eadc4a93b2366a7153a0b1fae67fcb"}

followthrough-ct-n128-r1 / seed 17301: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python",

"used_memory_mib": 1286}}, "index": 5, "memory_total_mib": 49140, "memory_used_mib": 1309, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ct-n128-r1/seed_17301/raw_samples.csv", "sha256": "efaeff45f89b0d17bfd9049b0e5c46e1923b530d122d92a48bc6193520c58d27", "size_bytes": 31468}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "6984cf8143eadc4a93b2366a7153a0b1fae67fcb"}

followthrough-ct-n512-r1 / seed 17300: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}, {"pid": 510483, "process_name": "/home/aidar.alimbayev/Documents/genmolv2.venv/bin/python", "used_memory_mib": 2434}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 3747, "name": "NVIDIA RTX A6000", "utilization_percent": 9, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT / raw-LOO", "parameterization": "raw_loo", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ct-n512-r1/seed_17300/raw_samples.csv", "sha256": "d4340e382079d41674f648c6c57f4fad70c57bd728eec942e8e076da2f1f6764", "size_bytes": 31379}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "6984cf8143eadc4a93b2366a7153a0b1fae67fcb"}

followthrough-ct-n512-r1 / seed 17301: {"checkpoint": "48986899c401c09cdc1e9e865e773cad40899b62129420689f89a8e622a9c99", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1256}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 1279, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "CT /

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followthrough-ce-n128-r1 / seed 17300: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 2}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ce-n128-r1/seed_17300/raw_samples.csv", "sha256": "12c37c23911a163d027ec5485cd8f6c2e0fa875ce0a9f155afe63945fc76d2f1", "size_bytes": 32757, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "6984cf8143eadc4a93b2366a7153a0b1fae67fcb", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

followthrough-ce-n128-r1 / seed 17301: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": 3}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ce-n128-r1/seed_17301/raw_samples.csv", "sha256": "5355cabcf13c8575dceb198f1e5168717cd70c4ad96f73dd67653fd7a291b0d5", "size_bytes": 31769, "sampling_budget": {"corrector_steps_per_molecule": 0,

"gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "6984cf8143eadc4a93b2366a7153a0b1fae67fcb", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

followthrough-ce-n512-r1 / seed 17300: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ce-n512-r1/seed_17300/raw_samples.csv", "sha256": "4cf561463c1dfd1ccbc69c7e493936f103dcb8356f26e17ab7f2421e37a6bee3", "size_bytes": 32455}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 512, "predictor_transitions_per_molecule": 512}, "source": "6984cf8143eadc4a93b2366a7153a0b1fae67fcb", "udlm_denoiser_metadata": {"inference_conversion": "subtract_local_forward_log_likelihood_before_controls", "objective": "clean_token_cross_entropy", "parameterization": "x0_denoiser", "schema_version": 1}}

followthrough-ce-n512-r1 / seed 17301: {"checkpoint": "b7d674ed5bddb1f597cb35b36cc6b64c0298eb28539d9cd01e720e7e46f6acc1", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 512, "parameterization": "x0_denoiser", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 1.0}, "denoiser_source_sha256": "344bf1801678a5cf9bd3e34894755afe4df0826f7f0475c03bcde6dd6a014f75", "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 512, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 512, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 1.0}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1564}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1587, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "objective": "clean CE", "parameterization": "x0_denoiser", "raw": {"relative_path": "output/udlm/followthrough_resolution_r1/followthrough-ce-n512-r1/seed_17301/raw_samples.csv", "sha256": "b103acd18cefb74fafd8e5b26ad928f328dc6bea7e85722e6e1ee1d15d9b0ed9", "size_bytes": 32286}, "sampling_budget": {"corrector_steps_per_molecule": 0,

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